

Project Details

Use the attached .sql file from BikeStore Database to carry out an analysis of BikeStore Company using the following guide questions.

Instructions

- Running queries and extracting data from database. Load the databases attached then connect them with the solution script.
- All answers should be completed directly on SSMS.

NB: Order Status 1= Pending

2= Processing

3= Rejected

4= Completed

Steps

A. Explore Data

Display all data on the tables

B. Questions

- 1) Which bike is most expensive? What could be the motive behind pricing this bike at the high price?
- 2) How many total customers does BikeStore have? Would you consider people with order status 3 as customers substantiate your answer?
- 3) How many stores does BikeStore have?
- 4) What is the total price spent per order?

Hint: total price = $[\text{list_price}] * [\text{quantity}] * (1 - [\text{discount}])$

- 5) What's the sales/revenue per store?

Hint: Sales revenue = $([\text{list_price}] * [\text{quantity}] * (1 - [\text{discount}]))$

- 6) Which category is most sold?
- 7) Which category rejected more orders?
- 8) Which bike is the least sold?
- 9) What's the full name of a customer with ID 259?
- 10) What did the customer on question 9 buy and when? What's the status of

this order?

- 11) Which staff processed the order of customer 259? And from which store?
- 12) How many staff does BikeStore have? Who seems to be the lead Staff at BikeStore?
- 13) Which brand is the most liked?
- 14) How many categories does BikeStore have, and which one is the least liked?
- 15) Which store still have more products of the most liked brand?
- 16) Which state is doing better in terms of sales?
- 17) What's the discounted price of product id 259?
- 18) What's the product name, quantity, price, category, model year and brand name of product number 44?
- 19) What's the zip code of CA?
- 20) How many states does BikeStore operate in?
- 21) How many bikes under the children category were sold in the last 8 months?
- 22) What's the shipped date for the order from customer 523
- 23) How many orders are still pending?
- 24) What's the names of category and brand does "Electra white water 3i - 2018" fall under?

25) Create a view that displays all completed orders with the following columns:

- Order ID
- Customer Full Name
- Store Name
- Staff Name
- Total Order Price

26) Create a view named `vw_ProductDetails` that contains:

- Product Name
- Brand Name
- Category Name
- Model Year
- List Price

View. ثم اعرض كل البيانات من الـ

27) Create a view that shows the total sales for each store.

Columns:

- Store Name
- Total Sales

28) Using a CTE, display customers who spent more than the average customer spending.

Columns:

- Customer ID
- Customer Name
- Total Spending

29) Using a CTE, rank products from highest to lowest revenue using `ROW_NUMBER()`.

Columns:

- Rank
 - Product Name
 - Revenue
-

30) Using a CTE, display the top 5 most sold bikes.

Columns:

- Product Name
 - Total Quantity Sold
-

31) Using a recursive CTE, display the staff hierarchy (Manager → Employee).

Columns:

- Staff Name
- Manager Name
- Level

(*staffs* في جدول *manager_id* يعتمد على وجود)

32) Find all products whose price is higher than the average product price using a subquery.

33) Find customers who have placed more orders than the average number of orders per customer.

34) Display the most expensive bike(s) using a subquery only.

35) Find all products that have never been ordered.

36) Find the store with the highest total sales using a subquery.

37) Display customers who purchased products from the most liked brand.

38) Find categories whose total sales are above the average category sales.

39) Display the products whose price is greater than the average price of their category.

40) Find the staff member who processed the highest number of completed orders.

Bonus (Advanced SQL)

41) Create a view named **vw_CustomerSales** showing:

- Customer ID
 - Customer Name
 - Number of Orders
 - Total Spending
-

42) Using the view created in Question 41, display customers whose total spending is greater than the average spending of all customers.

43) Using a CTE, calculate the cumulative sales by order date.

Columns:

- Order Date
 - Daily Sales
 - Running Total
-

44) Using a subquery, find the second most expensive bike.